

The minimax quadratic sign problem: exact values, oscillation superadditivity, and the two-scale barrier

A continuation of the study of MathOverflow question 413935

mo-413935-ai-attempt repository, second research note

1 August 2026

Abstract

For

$$F(n) = \min_{a_{ij} \in \{-1,1\}} \max_{x \in \{-1,1\}^n} \left| \sum_{1 \leq i < j \leq n} a_{ij} x_i x_j \right|,$$

the question whether $F(n)/n^{3/2}$ converges remains open. This note extends the July 2026 note in this repository. New content: the exact values $F(2), \dots, F(10) = 1, 3, 4, 4, 5, 9, 10, 12, 13$; a proof that the Gaussian-rounding method of the July note, fully optimized, computes exactly the oscillation functional $\text{Osc}_{1/2}(n) = \frac{1}{2} \min_A (\max_x S_A - \min_x S_A)$, which lies in $[F(n)/2, F(n)]$ and is superadditive; a merge theorem $F(n+m) \leq F(n) + F(m) + (1+o(1))n\sqrt{m}$ with a matching block floor of order $n\sqrt{m}$; a resulting continuity theory for $G(n) = F(n)/n^{3/2}$, including the statement that the limit exists if and only if it exists along conference orders; the observation that symmetric conference matrices attain their spectral ceiling exactly at orders $q+1$ with q an even prime power ($q=9, 25$ verified), so no uniform conference deficit exists; and an explicit bounded divergent profile satisfying every inequality proved here and in the July note, which shows that all such local inequalities together cannot decide convergence. The sharpest remaining target is a one-scale-to-all-scales amplification inequality. The question stays open.

1 The problem and what is new

Throughout, A is a symmetric matrix with zero diagonal and off-diagonal entries $a_{ij} \in \{-1, 1\}$, called a signing of K_n . Write

$$S_A(x) = \sum_{i < j} a_{ij} x_i x_j, \quad M(A) = \max_{x \in \{-1,1\}^n} |S_A(x)|, \quad F(n) = \min_A M(A), \quad G(n) = \frac{F(n)}{n^{3/2}}.$$

MathOverflow question 413935, posed by Paata Ivanisvili in January 2022, asks whether $\lim_n G(n)$ exists. The July 2026 note in this repository proves

$$\frac{1}{\pi} \leq \liminf G \leq \limsup G \leq \frac{1}{2}$$

and the covering-radius identity $F(n) = \binom{n}{2} - 2\rho(D_n)$. Those proofs were re-audited for the present note and stand; see `AUDIT.md`.

New results proved below.

1. Exact values $F(2), \dots, F(10)$ and the exact one-sided and oscillation optima for $n \leq 10$, none previously recorded (Section 2).
2. An even-odd inequality: for every signing, every vertex bipartition, and every pair (x, y) , $M \geq |S_P(x) + S_R(y)| + |x^\top C y|$, giving a floor of order $n\sqrt{m}$ for every two-block structure (Section 3).
3. A merge theorem $F(n + m) \leq F(n) + F(m) + \sqrt{nmq(n)}$ with $q(n) = (1 + o(1))n$, hence a modulus of continuity for G on multiplicative windows, the interval structure of the limit set, and the reduction of the limit question to conference orders (Section 4).
4. The oscillation functional $\text{Osc}(n)$ is superadditive, satisfies $F \leq \text{Osc} \leq 2F$, and equals the exact value of the July note's Gaussian-rounding method when that method is optimized over all correlation matrices; the constant $1/\pi$ is the value of the linear slice, and no polynomial slice can beat it on flat-spectrum signings (Section 5).
5. A fixed-temperature equivalence: for every fixed $\beta > 0$ the free-energy version of the problem differs from $G(n)$ by at most $\log 2/(\beta\sqrt{n})$, so the zero-temperature limit is harmless and the entire difficulty is the $n \rightarrow \infty$ interpolation (Section 6).
6. Relaxations collapse: couplings in $[-1, 1]^{\binom{n}{2}}$ give optimum 0; couplings on the Euclidean sphere of radius $\sqrt{\binom{n}{2}}$ give optimum exactly $\sqrt{\binom{n}{2}} = \Theta(n)$; near-optimal signings converge to the zero graphon (Section 7).
7. Symmetric conference matrices attain their spectral ceiling exactly at orders 10 and 26, and in general at any order $q + 1$, q an even power of an odd prime, whose switching class contains a suitable regular graph. The July note's "conference matrices do not attain their spectral ceiling" is therefore an artifact of $\sqrt{q} \notin \mathbb{Z}$ at the orders then checked (Section 8).
8. A consistency barrier: an explicit bounded divergent profile satisfies every inequality about the values $F(n)$ proved here and in the July note. Local estimates alone, however sharpened, cannot settle the question (Section 9).

Section 10 organizes three proof programs with their exact targets and failure points, and Section 11 states the sharpest remaining wall and the single recommended next lemma. All finite claims are reproducible from `verification/`; see the appendix.

2 Exact values and structural data

Proposition 2.1 (Parity). *Every value $S_A(x)$, hence $M(A)$ and $F(n)$, is congruent to $\binom{n}{2}$ modulo 2.*

Proof. $S_A(x)$ is a sum of $\binom{n}{2}$ odd terms. □

Theorem 2.2 (Exact values). *By exhaustive enumeration of switching classes (representatives with all edges at one vertex positive; the maximum is a switching-class invariant),*

n	2	3	4	5	6	7	8	9	10
$F(n)$	1	3	4	4	5	9	10	12	13
$G(n)$	0.354	0.577	0.500	0.358	0.340	0.486	0.442	0.444	0.411

Witness signings are recorded in `verification/data/results_2026-08.json` and reverified by an independent script.

Remarks on the data.

1. G is not monotone in either direction and oscillates through $n = 10$. The smallest ratios in the exact range occur at $n = 6$ and $n = 5$, the conference two-graph and its one-point deletion; among $7 \leq n \leq 10$ the smallest is the conference order $n = 10$.
2. At $n = 5, 6$ the optimal switching classes are exactly the conference-derived two-graphs: the Seidel spectra of the optimizers are $\{\pm\sqrt{5}^{(2)}, 0\}$ and $\{\pm\sqrt{5}^{(3)}\}$.
3. At $n = 10$ the optimum $F(10) = 13$ is strictly better than the conference class, whose maximum is 15 (Section 8), and the optimal spectrum $\{\pm\sqrt{17}, \pm 3.387^{(2)}, \pm 1.590^{(2)}\}$ is far from flat. Spectral flatness and minimax optimality diverge already at $n = 10$.
4. One-sided optimum. Enumeration gives $\min_A \max_x S_A(x) = \lfloor n/2 \rfloor$ for $3 \leq n \leq 10$, attained uniquely by the all-negative class. This matches the theorem of Petersdorf [1] that the maximum frustration index of a signing of K_n is $\lfloor (n-1)^2/4 \rfloor$, attained only by the all-negative class: via the cut-code dictionary the two statements are equivalent. The one-sided problem is therefore $\Theta(n)$, while the two-sided problem is $\Theta(n^{3/2})$: the absolute value is the entire content of the $n^{3/2}$ scale. In coding terms, the cut code has covering radius $\binom{n}{2}/2 - \Theta(n)$, while appending the all-ones word (the code D_n of the July note is the cut code extended by complementation, a linear code) drops the covering radius to $\binom{n}{2}/2 - \Theta(n^{3/2})$.
5. Certified upper bounds beyond the exact range, from principal submatrices of Paley conference matrices, from the exact tensor computation below, and from a beam extension search: $F(11) \leq 19$, $F(12) \leq 18$, $F(13) \leq 20$, $F(14) \leq 21$, $F(15) \leq 29$, $F(16) \leq 32$, $F(17) \leq 32$, $F(20) \leq 44$.

3 The even-odd inequality and block floors

Lemma 3.1 (Even-odd split). *Let B be a signing of K_{n+m} , let (P, R) be a bipartition of the vertices with $|P| = n$, $|R| = m$, with induced signings A on P , A' on R and cross block $C \in \{-1, 1\}^{n \times m}$. Then for all $x \in \{-1, 1\}^P$, $y \in \{-1, 1\}^R$,*

$$M(B) \geq |S_A(x) + S_{A'}(y)| + |x^\top C y|.$$

Proof. $S_B(x \oplus y) = e + c$ and $S_B((-x) \oplus y) = e - c$ with $e = S_A(x) + S_{A'}(y)$ and $c = x^\top C y$, because S_A is even in x while the cross term is odd. Now $\max(|e + c|, |e - c|) = |e| + |c|$. $\square$

Corollary 3.2 (Block floor). *For every signing B of K_{n+m} and every bipartition as above,*

$$M(B) \geq \max_{x, y} |x^\top C y| \geq \max(n\mu_m, m\mu_n), \quad \mu_k := \mathbb{E} \left| \sum_{i=1}^k \varepsilon_i \right|,$$

where ε_i are independent uniform signs. Since $\mu_k \geq \sqrt{k/3}$ and $\mu_k = \sqrt{2k/\pi} (1 + O(1/k))$, every two-block construction on $n + m$ vertices pays a cross cost of order $n\sqrt{m}$, and for balanced blocks a cost at least $(\sqrt{2/\pi} - o(1))(N/2)^{3/2} = (0.28\dots)N^{3/2}$, which is at the target scale. Optimizing the bipartition size reproduces the July note's bipartite-exposure bound $\liminf G \geq \frac{2}{3}\sqrt{2/(3\pi)} \approx 0.3071$.

Proof. Take y uniform and $x = \text{sgn}(Cy)$ coordinatewise: $\mathbb{E}_y \|Cy\|_1 = \sum_{i \in P} \mathbb{E} |\langle c_i, y \rangle| = n\mu_m$. The bound $\mu_k \geq k^{3/2}/\sqrt{\mathbb{E}S_k^4} \geq \sqrt{k/3}$ follows from Hölder with $\mathbb{E}S_k^2 = k$, $\mathbb{E}S_k^4 = 3k^2 - 2k$. The rest is Lemma 3.1 with the even part discarded. $\square$

Remark 3.3. The even part cannot currently be exploited: at the maximizing (x, y) of the cross term, the designer of B can arrange the block energies to be $O(n)$, since block energies at a fixed pair of points have no lower bound at scale $n^{3/2}$. Converting the additive $|e|$ term into a gain requires anticoncentration of block energies at rounded points, which is exactly the missing landscape control discussed in Section 10.

4 Merging, continuity, and subsequence reductions

Let $q(n)$ denote the least prime $q \equiv 1 \pmod{4}$ with $q \geq n - 1$. By the prime number theorem for arithmetic progressions, $q(n) = (1 + o(1))n$.

Theorem 4.1 (Merge). *For $1 \leq m \leq n$,*

$$F(n + m) \leq F(n) + F(m) + \sqrt{nmq(n)} = F(n) + F(m) + (1 + o(1))n\sqrt{m}.$$

Proof. Join optimal signings A_n^*, A_m^* by a cross block C equal to any $n \times m$ submatrix of a symmetric conference matrix of order $q(n) + 1$; then $\|C\|_{\text{op}} \leq \sqrt{q(n)}$ and $|x^\top Cy| \leq \sqrt{nm} \|C\|_{\text{op}}$. Apply the triangle inequality to $S = S_{A_n^*} + S_{A_m^*} + x^\top Cy$. $\square$

By Corollary 3.2, no choice of cross block can beat this by more than the constant: the cross cost of any two-block construction is between $(\sqrt{2/\pi} - o(1))n\sqrt{m}$ and $(1 + o(1))n\sqrt{m}$. Block composition is thus exactly a $\Theta(n\sqrt{m})$ mechanism: cheap only for $m = o(n)$, and structurally incapable of reaching balanced scales, in agreement with the wall recorded in the July note.

Theorem 4.2 (Continuity on multiplicative windows). *Let $\bar{G} = \sup_k G(k)$, which is finite. For every $\varepsilon \in (0, 1]$,*

$$\limsup_{n \rightarrow \infty} \sup_{n \leq n' \leq (1+\varepsilon)n} |G(n') - G(n)| \leq \left(1 + \frac{5}{2} \bar{G}\right) \sqrt{\varepsilon}.$$

Proof. Write $n' = n + m$, $m \leq \varepsilon n$. Upward: by Theorem 4.1, $G(n') \leq [G(n) + G(m)(m/n)^{3/2} + (1 + o(1))\sqrt{m/n}](1 + m/n)^{-3/2} \leq G(n) + \bar{G}\varepsilon^{3/2} + (1 + o(1))\sqrt{\varepsilon}$. Downward: F is nondecreasing (restriction of a signing does not increase the maximum, by averaging over the last sign), so $G(n') \geq G(n)(n/n')^{3/2} \geq G(n)(1 - \frac{3}{2}\varepsilon)$, giving $G(n) - G(n') \leq \frac{3}{2}\bar{G}\varepsilon$. Combine, using $\varepsilon \leq \sqrt{\varepsilon}$. $\square$

Corollary 4.3 (Interval limit set). *$|G(n+1) - G(n)| \rightarrow 0$, so the set of limit points of $(G(n))$ is a closed interval $[L^-, L^+] \subseteq [1/\pi, 1/2]$. Convergence fails if and only if $L^- < L^+$, in which case every value in (L^-, L^+) is hit infinitely often.*

Corollary 4.4 (Log-dense subsequences suffice). *Let (n_k) be any sequence with $n_{k+1}/n_k \rightarrow 1$, for example the conference orders $\{q+1 : q \text{ prime}, q \equiv 1 \pmod{4}\}$. If $G(n_k)$ converges, then $G(n)$ converges to the same limit. In particular the limit exists if and only if it exists along conference orders.*

Proof. Bracket $n \in [n_k, n_{k+1}]$ and apply Theorem 4.2 with $\varepsilon_k = n_{k+1}/n_k - 1 \rightarrow 0$. For the conference orders, gaps between consecutive primes $\equiv 1 \pmod{4}$ are $o(x)$ by the prime number theorem in arithmetic progressions. $\square$

Corollary 4.5 (Amplification criterion). *Suppose there are $n_j \rightarrow \infty$ with $G(n_j) \rightarrow L := \liminf G$ and $\eta(n_j) \rightarrow 0$ such that*

$$F(N) \leq (N/n_j)^{3/2} F(n_j) + \eta(n_j) N^{3/2} \quad \text{for all } N \geq n_j.$$

Then $\lim G = L$. The same holds with $\liminf$ replaced by $\limsup$ and the inequality reversed for restrictions.

Proof. $G(N) \leq G(n_j) + \eta(n_j)$ for all $N \geq n_j$; let $N \rightarrow \infty$, then $j \rightarrow \infty$. $\square$

The criterion demands control from one scale to all larger scales. Section 9 shows that this uniformity cannot be weakened to comparisons at any fixed ratio.

5 The oscillation functional

Define, for a signing A and for the minimax problem,

$$\text{osc}(A) = \max_x S_A(x) - \min_x S_A(x), \quad \text{Osc}(n) = \min_A \text{osc}(A), \quad \text{Osc}_{1/2}(n) = \frac{1}{2} \text{Osc}(n).$$

Proposition 5.1. *For every A : $\frac{1}{2} \text{osc}(A) \leq M(A) \leq \text{osc}(A)$, hence $F(n)/2 \leq \text{Osc}_{1/2}(n) \leq F(n)$.*

Proof. $\sum_x S_A(x) = 0$, so $\max S \geq 0 \geq \min S$; then $M = \max(\max S, -\min S)$ lies between the average and the sum of the two nonnegative numbers $\max S$ and $-\min S$. For the right-hand inequality over minima, evaluate osc at an F -optimal signing. $\square$

Theorem 5.2 (Superadditivity). *For every signing B of K_{n+m} and every bipartition (P, R) , $\text{osc}(B) \geq \text{osc}(B|_P) + \text{osc}(B|_R)$. Consequently*

$$\text{Osc}(n+m) \geq \text{Osc}(n) + \text{Osc}(m) \quad \text{and} \quad F(n+m) \geq \text{Osc}_{1/2}(n) + \text{Osc}_{1/2}(m).$$

Proof. Let x^+ maximize $S_{B|_P}$ and y^+ maximize $S_{B|_R}$. Since the cross term is odd under $x \mapsto -x$ while both block terms are even, one of $(\pm x^+, y^+)$ has cross term ≥ 0 , so $\max S_B \geq \max S_{B|_P} + \max S_{B|_R}$. Symmetrically $\min S_B \leq \min S_{B|_P} + \min S_{B|_R}$. Subtract. The second display follows by evaluating at optimal signings and using Proposition 5.1. $\square$

Theorem 5.3 (Exact value of the Gaussian-rounding method). *For correlation matrices $R^\pm$ (unit diagonal, positive semidefinite) and arcsin applied entrywise,*

$$M(A) \geq \frac{1}{2\pi} \langle A, \arcsin R^+ - \arcsin R^- \rangle,$$

and the supremum of the right side over all pairs (R^+, R^-) equals $\frac{1}{2}(\max_x S_A - \min_x S_A)$. Hence the strongest bound obtainable from pairs of Gaussian sign roundings is exactly $F(n) \geq \text{Osc}_{1/2}(n)$, and by Proposition 5.1 this method loses at most a factor 2 and is exact whenever some optimal signing has symmetric spread.

Proof. Realize R as the correlation matrix of a Gaussian vector z , set $x = \text{sgn}(z)$, and use $\mathbb{E}[\text{sgn } z_i \text{sgn } z_j] = \frac{2}{\pi} \arcsin R_{ij}$: then $\mathbb{E} S_A(x) = \frac{1}{\pi} \langle A, \arcsin R \rangle$ up to the $i < j$ normalization, and $|\mathbb{E} S_A| \leq M(A)$; subtract the two choices. For the supremum: any candidate value is $\frac{1}{2}(\mathbb{E}^+ S_A - \mathbb{E}^- S_A) \leq \frac{1}{2}(\max S - \min S)$ since each expectation lies in $[\min S, \max S]$; the rank-one choices $R^\pm = x^\pm (x^\pm)^\top$ with x^+ maximizing and x^- minimizing S_A give $\arcsin R^\pm = \frac{\pi}{2} x^\pm (x^\pm)^\top$ and attain it. $\square$

Proposition 5.4 (The linear slice and its optimality). *The July note's choice $R^\pm = \text{corr}((I \pm tA/\sqrt{n})g)$ with the optimal t yields $F(n) \geq n\sqrt{n-1}/\pi$. More generally, for any symmetric W with zero diagonal and $(W^2)_{ii}$ constant in i , the slice $R^\pm = \text{corr}((I \pm W)g)$ has linear-response value*

$$\frac{2}{\pi} \cdot \frac{\text{tr}(AW)}{1 + \frac{1}{n} \text{tr}(W^2)} \leq \frac{2}{\pi} \cdot \frac{\|A\|_F \|W\|_F}{1 + \frac{1}{n} \|W\|_F^2} \leq \frac{n\sqrt{n-1}}{\pi},$$

by Cauchy–Schwarz, $\|A\|_F^2 = n(n-1)$, and $\sigma/(1+\sigma^2) \leq \frac{1}{2}$ with $\sigma^2 = \|W\|_F^2/n$, with equality only for $W \propto A$. In particular no matrix function of A , and no other spectrally defined choice of W , improves the constant $1/\pi$ at this order. The curvature corrections from arcsin are nonnegative, and for quadratic slices they are controlled by the spectral excess $X(A) = \sum_i \lambda_i^4/(n(n-1)^2) - 1 \geq 0$: they vanish exactly on conference matrices, via $\langle A, AB + BA \rangle = 2 \text{tr}(A^2 B) = 2n(n-1)^2 X(A)$ for $B = A^2 - (n-1)I$.

Proof. The first claim is the July note's Theorem 2, reverified. For the second, expand $\Sigma^\pm = (I \pm W)^2 = I \pm 2W + W^2$, whose diagonal $d = 1 + (W^2)_{ii}$ is constant by hypothesis, so $\langle A, R^+ - R^- \rangle = 4 \text{tr}(AW)/d$ (the diagonal of A vanishes). When W is sign-aligned with A (that is, $a_{ij}W_{ij} \geq 0$ for all $i < j$, as for $W = tA/\sqrt{n}$), the per-edge bound $\arcsin(u+v) - \arcsin(u-v) \geq 2v$ of the July note applies with $v_{ij} = 2a_{ij}W_{ij}/d \geq 0$ and certifies the displayed value; without sign alignment the per-edge bound degrades on the misaligned edges, so the certified value is no larger. In either case Cauchy–Schwarz bounds the display by $n\sqrt{n-1}/\pi$. The trace identity for the curvature term is a direct computation. $\square$

Remark 5.5 (Why this matters). Three consequences. First, $1/\pi$ is not a ceiling of the Gaussian-rounding method; it is the value of its simplest slice. The full method computes $\text{Osc}_{1/2}(n)$, which the data below place at $F(n) - O(1)$ for $n \leq 10$. Second, the adversary against every polynomial slice is a flat-spectrum signing, and actual flat-spectrum signings (conference matrices) have true value far above $1/\pi$ (Section 8); the loss happens because polynomial slices cannot see Boolean structure, only spectra. Third, replacing $\arcsin R$ by R (the semidefinite relaxation) is provably lossy: the Grothendieck-type gap on K_n is $\Theta(\log n)$ [2, 3], so any argument through the linearized functional must lose a $\log n$ factor in general; the arcsine nonlinearity, or the restriction to the minimizing A , must be used.

Enumeration gives the oscillation optima

n	3	4	5	6	7	8	9	10
$\text{Osc}(n)$	4	8	8	10	16	20	24	26
$2F(n)$	6	8	8	10	18	20	24	26

so $\text{Osc} = 2F$ whenever parity permits and $\text{Osc} = 2F - 2$ otherwise ($n = 3, 7$). The superadditivity of Theorem 5.2 is strict in the data.

Conjecture 5.6 (Symmetric spread). $\text{Osc}_{1/2}(n) = F(n) - O(n)$; equivalently, some near-optimal signings have $\max S \approx -\min S$. Under this conjecture F is asymptotically superadditive: $F(n+m) \geq F(n) + F(m) - O(n)$.

Even granting Conjecture 5.6, superadditivity does not force convergence at the superextensive scale $n^{3/2}$: summing k blocks of size n captures only $kn^{3/2}$ of the $(kn)^{3/2} = k^{3/2}n^{3/2}$ target, so Fekete-type arguments are vacuous in both directions. This is quantified in Section 9.

6 Fixed-temperature equivalence

Proposition 6.1. *For $\beta > 0$ let $\Phi_{n,\beta} = \min_A \frac{1}{\beta n^{3/2}} \log \sum_{x \in \{-1,1\}^n} e^{\beta |S_A(x)|}$. Then*

$$0 \leq \Phi_{n,\beta} - G(n) \leq \frac{\log 2}{\beta \sqrt{n}}.$$

Hence for every fixed $\beta > 0$ (indeed whenever $\beta_n \sqrt{n} \rightarrow \infty$), $\liminf$, $\limsup$, and existence of the limit coincide for $\Phi_{n,\beta}$ and $G(n)$. The same holds for the sign-augmented partition function $\sum_{x,s} e^{\beta s S_A(x)}$ with error $\log(2^{n+1})/(\beta n^{3/2})$.

Proof. $e^{\beta M} \leq \sum_x e^{\beta |S|} \leq 2^n e^{\beta M}$ for each A ; take log, divide, minimize. $\square$

Two of the concerns listed against the finite-temperature route are therefore empty: no uniformity in β is needed, and the absolute value can be handled exactly. What remains open is precisely the $n \rightarrow \infty$ behavior at fixed β with adversarial couplings. The July note's annealed obstruction is confirmed quantitatively: for a balanced merge the annealed cross cost is $nm \log \cosh \beta / (\beta N^{3/2}) \asymp \beta N^{1/2}$, which requires $\beta \sqrt{N} \rightarrow 0$, while the sandwich above requires $\beta \sqrt{N} \rightarrow \infty$. The window is empty, and this is structural for annealing, not an artifact.

7 Relaxations collapse

Proposition 7.1 (Box and sphere). *Let $m = \binom{n}{2}$ and let $\chi(x) = (x_i x_j)_{i < j}$.*

1. $\min_{a \in [-1,1]^m} \max_x |\langle a, \chi(x) \rangle| = 0$, attained at $a = 0$; every convex relaxation of the coupling cube whose feasible set contains 0 has optimum 0.
2. $\min_{\|a\|_2 = \sqrt{m}} \max_x |\langle a, \chi(x) \rangle| = \sqrt{m}$, attained by concentrating all mass on a single edge: $|\langle a, \chi \rangle| \equiv \sqrt{m}$ for $a = \sqrt{m} e_{\{1,2\}}$. The lower bound is $\max_x \langle a, \chi \rangle^2 \geq \mathbb{E}_x \langle a, \chi \rangle^2 = \|a\|_2^2$.

So the ± 1 coupling constraint plays two distinct roles at once: it fixes the total mass and it forces the mass to spread over all m edges. Dropping either role collapses the value from $\Theta(n^{3/2})$ to $O(n)$. The integrality gap of any relaxation of this kind is the entire phenomenon, and rounding a relaxed optimizer back to signs must inject, not preserve, the $n^{3/2}$ scale. This disposes of the relaxed-couplings attack in its natural formulations.

Theorem 7.2 (Zero graphon). *If $M(A_n) = O(n^{3/2})$, then the cut norm satisfies $\|A_n\|_{\square} \leq 4M(A_n) = O(n^{3/2}) = o(n^2)$: every sequence of near-optimal signings converges to the zero graphon, and every functional continuous in the dense graph (cut-metric) topology is blind to the problem. Any limit theory that could express $\lim G$ must live at the $n^{3/2}$ fluctuation scale, for which no compactness theorem is currently available.*

Proof. $\|A\|_{\square} \leq \max_{x,y \in \{-1,1\}^n} |x^\top A y|$ by writing indicators as $(1+x)/2$; polarization and multilinearity of the zero-diagonal quadratic form give $\max_{x,y} |x^\top A y| \leq 2 \max_z |z^\top A z| = 4M(A)$. $\square$

8 Conference matrices at their ceiling

Theorem 8.1 (Attainment). *Let C be a symmetric conference matrix of order N . Then $M(C) \leq \frac{1}{2} N \sqrt{N-1}$, with equality if and only if some $x \in \{-1,1\}^N$ is an eigenvector of C , necessarily*

with eigenvalue $\pm\sqrt{N-1} \in \mathbb{Z}$. Equality holds at $N = 10$ (Petersen two-graph, $x = \mathbf{1}$ for the Petersen-form Seidel matrix, value 15) and at $N = 26$ (Paley over $\mathbb{F}_{25}$, value 65), both verified exactly. Equality is impossible whenever $N - 1$ is not a perfect square, in particular at the orders 6, 14, 18, 30 checked in the July note, where the exact values are 5, 21, 33, 75 and the normalized ratios 0.340, 0.401, 0.432, 0.456 increase toward $1/2$.

Proof. Cauchy–Schwarz: $|x^\top Cx| \leq \|x\| \|Cx\| = N\sqrt{N-1}$, with equality iff $Cx \parallel x$; integrality of $x^\top Cx$ forces $\sqrt{N-1} \in \mathbb{Q}$. A ± 1 eigenvector with eigenvalue $\sqrt{N-1}$ exists iff the switching class contains a graph that is regular of degree $(N-1-\sqrt{N-1})/2$ (switch by the eigenvector and read off constant row sums): degree 3 at $N = 10$ (Petersen), degree 10 at $N = 26$ (a Paulus graph). The stated values are exhaustive computations with recorded witnesses. $\square$

Consequences.

1. There is no uniform conference deficit. The July note’s wall “small Paley examples do not attain the spectral ceiling” is explained entirely by $\sqrt{N-1} \notin \mathbb{Z}$ at those orders. At orders $q+1$ with q an even power of an odd prime, whenever the switching class contains a regular graph of degree $(q-\sqrt{q})/2$, the construction value is exactly $\frac{1}{2}(q+1)\sqrt{q}$, of normalized ratio $\frac{1}{2}\sqrt{q/(q+1)} \rightarrow \frac{1}{2}$. Candidate orders 10, 26, 50, 82, $\dots$; whether infinitely many contain such regular descendants is a known open-ended question in the theory of regular two-graphs.
2. Improving $\limsup G \leq \frac{1}{2}$ therefore requires either non-conference matrices or a proof that the minimum beats the conference class at scale. The datapoint $F(10) = 13 < 15$ shows the minimum does beat the conference class at $n = 10$, by a non-flat signing. The upper-bound program should not fixate on spectrally flat matrices.
3. The certification barrier remains: for a random signing the true maximum is $(0.7632\dots)n^{3/2}$ (the Parisi ground state, universal for ± 1 couplings [4, 5]), while no known efficient certificate goes below the spectral $n^{3/2}$; and for structured matrices every known certificate below the spectral value is exhausted by exact enumeration. New certification ideas (Krawtchouk or Delsarte arguments on the coset structure of the extended cut code) are listed as an open direction, with the caution that the code’s dual weight spectrum is too rich for the standard external-distance bound.

9 The consistency barrier

Everything proved so far constrains G only locally in scale. The next proposition shows this is a hard limitation of the entire inequality set, not of the sharpness of individual constants.

Proposition 9.1 (A divergent profile consistent with all proved inequalities). *Define*

$$\tilde{G}(n) = \frac{2}{5} + \frac{1}{20} \sin(2\pi \log_2 \log_2 n) \quad (n \geq 4), \quad \tilde{F}(n) = \tilde{G}(n) n^{3/2}.$$

Then, for all sufficiently large n and all $1 \leq m \leq n$: $\tilde{F}$ is strictly increasing; $\tilde{F}(n+1) \leq \tilde{F}(n) + n$; $\tilde{F}(n+m) \leq \tilde{F}(n) + \tilde{F}(m) + 1.01 n\sqrt{m}$; $\tilde{F}(n+m) \geq \tilde{F}(n) + \tilde{F}(m)$; $\tilde{F}(n)/n^{3/2} \in [1/\pi, 1/2]$; and $\tilde{G}$ satisfies the conclusions of Theorems 4.2 and Corollaries 4.3, 4.4 vacuously or with margin. Yet $\tilde{G}(n)$ diverges: its limit set is $[0.35, 0.45]$.

Proof. All verifications are elementary calculus from $\tilde{G}'(n) = O(1/(n \log n))$, so that $\tilde{G}$ is asymptotically constant on any window of bounded ratio, together with $\frac{d}{dn}\tilde{F} = n^{1/2}(\frac{3}{2}\tilde{G} + o(1)) \in [0.5 n^{1/2}, 0.7 n^{1/2}]$. Monotonicity and the Lipschitz bound follow at once. For the merge inequality, the left side minus $\tilde{F}(n)$ is $\frac{3}{2}\tilde{G}\varepsilon n^{3/2}(1+o(1))$ with $\varepsilon = m/n$, which is at most $\tilde{G}(m)m^{3/2} + 1.01 n\sqrt{m}$ because $0.7\varepsilon \leq 1.01\sqrt{\varepsilon}$; superadditivity holds because $0.5\varepsilon n^{3/2} \geq 0.45\varepsilon^{3/2}n^{3/2}$ for $\varepsilon \leq 1$. The oscillation of $\log_2 \log_2 n$ is unbounded while its increment over $[n, 2n]$ is $O(1/\log n)$, so the limit set is the full interval. Numerical verification of every inequality with explicit margins for $n \leq 10^6$ is included in the verification suite. $\square$

The profile also satisfies the oscillation chain of Section 5 with $\widetilde{\text{Osc}} := 2\tilde{F}$, and Conjecture 5.6. Conclusion: the union of all inequalities proved in this repository, even sharpened by constants, cannot decide the question. Deciding it requires an inequality that couples scale n to scales N with $N/n \rightarrow \infty$, such as the amplification criterion of Corollary 4.5. Comparisons at any fixed ratio λ , for instance a proof that $G(2n) - G(n) \rightarrow 0$, would still not suffice: $\tilde{G}$ satisfies that too.

10 Three proof programs

Program 1: ellipsope slices (lower bound)

Target lemma: a map $A \mapsto (R^+(A), R^-(A))$ into correlation-matrix pairs, analyzable uniformly in A , with $\frac{1}{2\pi}\langle A, \arcsin R^+ - \arcsin R^- \rangle \geq (1/\pi + \delta) n^{3/2}$ for some $\delta > 0$. By Theorem 5.3 the program is complete in principle: the full supremum equals $\text{Osc}_{1/2}(n) \geq F(n)/2$, and empirically $\text{Osc}_{1/2} = F - O(1)$. Failure points identified: every polynomial slice degenerates on flat-spectrum signings (Proposition 5.4); rank-one slices require knowing extremal Boolean vectors, which is circular; the linearized functional is off by $\Theta(\log n)$ in general [2]. Repair directions: mixtures of a polynomial slice with low-rank Boolean-structured corrections, controlled by an anticoncentration statement for block energies at rounded points (the missing landscape lemma), or slices adapted to the bipartite exposure structure of Lemma 3.1 (Rademacher rather than Gaussian roundings). The obstruction appears structural for slices defined by spectral data alone, and technical, possibly movable, for Boolean-aware slices.

Program 2: amplification (upper bound)

Target lemma (Corollary 4.5): from one good signing at size n , signings at all $N \geq \Lambda(n)$ with $G(N) \leq G(n) + o_n(1)$. Exact losses of the known lifts: block merges cost $\Theta(n\sqrt{m})$ exactly (Theorem 4.1 and Corollary 3.2), fatal at balanced scales; iterated one-vertex extension is far weaker in practice (beam search from the exact $F(10)$ optimizer reaches only ratio ≈ 0.52 by $n = 14$, worse than conference restrictions); tensor doubling $A \otimes H_2$ (intra-pair edges refilled) does not preserve the constant: exact values move the ratios $0.358 \rightarrow 0.411$, $0.340 \rightarrow 0.433$, $0.486 \rightarrow 0.477$, $0.442 \rightarrow 0.500$, $0.411 \rightarrow 0.492$ at $n = 5, 6, 7, 8, 10$. The data are consistent with iterated tensoring converging to its own fixed constant near $1/2$ rather than preserving the base ratio: misaligned block vectors let the adversary realize Gram-relaxation values, and the loss is a Grothendieck-type constant, not $o(1)$. Repair directions: lifts along structured algebraic extensions (subfield towers of Paley matrices, where the added symmetry may rule out misaligned adversaries), or merge trees whose cross blocks are chosen adaptively against the blocks' own near-maximizers rather than obliviously. The oblivious version of the obstruction is proved (Corollary 3.2); whether adaptive cross blocks can beat $n\sqrt{m}$ is open and is the sharpest open sub-question of this program.

Program 3: adversarial thermodynamic interpolation

Target: at fixed β , an interpolation between $\Phi_{N,\beta}$ and block free energies with normalized cost $o(1)$. Status: the sandwich of Proposition 6.1 makes the temperature exchange exact; the annealed route has an empty parameter window (verified quantitatively in Section 6); the Guerra–Toninelli mechanism [5] needs a disorder average and Gaussian integration by parts to produce a signed overlap term, and the minimum over couplings supplies neither: the derivative of a minimum runs through the argmin, whose structure is exactly what is unknown. A quenched interpolation with adversarial couplings would need a new convexity input in the coupling variable. Note that the natural convexity is absent: $A \mapsto \log \sum e^{\beta|S_A|}$ is convex, so its minimum over the cube is attained at vertices (consistent with the collapse of Proposition 7.1), but minima of convex functions do not interpolate. The obstruction appears structural.

11 The sharpest wall and the next lemma

The problem is now reduced, with proofs, to the following statement. All local behavior of G is understood at the level needed: G has an interval limit set inside $[1/\pi, 1/2]$, is continuous on bounded multiplicative windows, and its limit exists iff it exists along conference orders. Every additive composition mechanism is exactly priced: merges cost $\Theta(n\sqrt{m})$, oscillation is superadditive, and the superextensive scale makes both Fekete directions vacuous. The value class of all proved inequalities admits divergent profiles (Proposition 9.1), so no refinement of these local estimates decides the question.

What is missing is a single scale-coupling inequality. The recommended target, the most constrained version consistent with everything above:

Conjecture 11.1 (Amplification lemma). *There is a function η with $\eta(n) \rightarrow 0$ such that for every n and every $N \geq n^2$,*

$$F(N) \leq (N/n)^{3/2} F(n) + \eta(n) N^{3/2}.$$

By Corollary 4.5 this implies convergence. The exponent 2 in $N \geq n^2$ is inessential; any threshold $\Lambda(n) < \infty$ works. The evidence: tensor and merge lifts fail by constants, not by scale, and the failure constant shrinks as the base ratio approaches the tensor fixed point; the conjecture only requires lossless amplification from optimal bases, whose structure (exact witnesses above) is rigid and highly symmetric at small n . The first concrete test: construct, from the exact $F(10)$ optimizer or the order-26 conference class, signings at $n \approx 10^2$ with ratio provably below 0.46. A disproof of Conjecture 11.1 in a strong form (a lower-bound mechanism active only at special scales) is the only currently visible route to nonconvergence, and nothing in the data suggests such a mechanism: the upper envelope moves smoothly, and all known algebraic families exist at log-dense orders, which Corollary 4.4 then smooths out.

A Computational protocol

All computations are exhaustive or witness-carrying; none are heuristic except the beam search, which only produces upper bounds that are then verified exactly.

- **exact_fn.c**: exact $F(n)$, one-sided optimum, and $\text{Osc}(n)$ for $n \leq 10$ by Gray-code enumeration of all switching classes (normalized representatives), with periodic from-scratch recomputation and a final independent audit of reported minima. $n = 10$ (about $6.9 \cdot 10^{10}$ class evaluations) was run as 8 disjoint fixed-prefix slices, each auditing its own minima.

- `conference_max.c`: symmetric conference matrices at orders 6, 10, 14, 18, 26, 30 (Paley over prime fields, Petersen Seidel form, Paley over $\mathbb{F}_{25}$), verified to satisfy $C^2 = (N - 1)I$, exact Boolean maxima with witnesses, signed spreads, and nested restriction chains inside order 30.
- `ext_chain.c`: beam extension upper bounds from the exact $F(10)$ witness.
- `verify_new_results.py`: independent reverification of every table in this note from recorded witnesses, full independent enumeration for $n \leq 6$, parity checks, Petersdorf check, tensor recomputation, the finite merge instance, numerical margins for Proposition 9.1 up to $n = 10^6$, and corruption controls that must fail when data are perturbed.

Restriction-chain data (order 30, three random nested chains): normalized ratios decrease monotonically in the subgraph size from about 0.7 at $k = 8$ to 0.456 at $k = 30$. Restrictions of good large signings are relatively poor small signings; the naive restriction-decay route to the reversed amplification criterion fails on conference matrices.

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